

EDINBURGH GOGARBANK, United Kingdom

WMO: 031660

Lat: 55.9283N

Lon: 3.3444W

Elev: 187

StdP: 14.60

Time Zone: 0.00 (EUW)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	27.6	20.6	15.5	28.5	23.0	17.4	30.8	32.9	45.7	29.8	47.0	3.2	230	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	72.5	62.6	69.6	60.8	67.2	59.5	64.2	70.2	62.4	67.4	60.9	65.3	9.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	82.3	66.8	60.2	78.5	65.2	58.7	74.3	63.3	29.4	70.2	28.2	67.5	27.1	65.4	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.0	22.9	20.3	20.1	78.9	4.4	4.2	16.9	82.0	14.3	84.4	11.8	86.8	8.6	89.9		
(5)			19.5	67.8	4.4	2.4	16.3	69.6	13.7	71.0	11.2	72.3	8.0	74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	39.8	40.2	42.4	46.7	51.1	56.2	59.4	58.6	55.5	49.8	43.6	40.2		
	DBStd	8.65	5.59	5.22	5.13	4.62	4.60	3.71	3.79	3.26	4.04	4.86	5.58	6.80		
	HDD50	1569	317	276	242	117	40	1	0	0	5	63	201	307		
	HDD65	5966	781	694	701	548	430	265	181	199	286	470	642	769		
	CDD50	1086	1	2	6	19	75	187	290	267	170	58	8	3		
	CDD65	9	0	0	0	0	0	1	6	1	1	0	0	0		
	CDH74	62	0	0	0	0	4	7	46	4	1	0	0	0		
	CDH80	4	0	0	0	0	0	0	4	0	0	0	0	0		
	WSAvg	9.2	10.7	9.7	10.5	9.7	9.1	8.0	7.6	8.0	9.0	8.8	9.4	10.2		
	PrecAvg	26.0	2.4	1.7	1.9	1.6	2.0	2.0	2.2	2.3	2.5	2.7	2.4	2.3		
(15) Precipitation	PrecMax	33.4	4.7	4.4	3.6	3.7	5.4	5.2	6.1	4.7	5.9	5.2	7.8	5.1		
	PrecMin	17.2	0.7	0.1	0.6	0.2	0.4	0.2	0.2	0.2	0.3	0.6	0.4	0.4		
	PrecStd	4.0	1.1	1.1	0.9	0.8	1.1	1.2	1.3	1.2	1.5	1.3	1.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	55.8	59.8	65.4	71.6	74.2	79.2	73.0	71.0	63.7	58.7	56.1		
		MCWB	51.4	50.4	51.5	55.0	61.0	63.5	65.7	63.3	61.9	58.0	55.0	52.8		
	2%	DB	51.9	52.7	55.2	60.2	66.0	69.7	73.8	69.7	67.4	61.1	56.2	54.1		
		MCWB	49.4	49.4	49.9	52.0	56.9	60.8	63.7	61.2	59.7	56.6	53.1	51.2		
	5%	DB	50.1	50.7	52.6	57.5	62.1	66.6	70.4	67.5	64.9	59.3	54.0	52.1		
		MCWB	47.4	47.6	48.1	50.4	54.5	58.7	61.6	59.7	58.4	55.4	51.0	49.5		
	10%	DB	48.4	48.6	50.4	55.1	59.3	64.0	67.3	65.6	62.8	57.4	52.0	50.1		
		MCWB	45.8	45.6	46.4	48.9	52.6	56.8	59.9	58.6	57.1	53.5	49.1	47.5		
	0.4%	WB	51.8	52.1	53.4	56.1	62.5	65.1	67.5	65.3	63.8	59.4	55.8	53.1		
		MCDB	53.7	53.9	57.9	62.6	70.1	71.4	77.6	71.3	69.5	62.1	57.9	55.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.7	49.6	50.8	53.4	58.2	61.7	64.8	62.9	61.3	57.5	53.4	51.4		
		MCDB	51.7	52.2	54.4	59.4	63.7	67.7	71.8	67.2	65.4	60.4	55.7	53.7		
	5%	WB	47.7	47.8	48.6	51.2	55.7	59.5	62.9	61.5	59.5	55.8	51.4	49.7		
		MCDB	49.8	50.5	52.0	56.4	60.4	64.9	68.5	65.5	63.6	58.7	53.7	52.1		
	10%	WB	45.9	45.8	46.7	49.5	53.7	57.8	61.1	60.1	57.9	54.1	49.3	47.6		
		MCDB	48.3	48.4	50.1	54.1	58.2	62.8	65.8	63.9	61.5	57.1	51.6	49.9		
	5% DB	MDBR	8.5	9.7	11.2	13.1	14.1	13.7	14.4	13.6	12.8	11.5	10.0	9.4		
		MCDBR	9.9	10.8	13.6	17.5	18.4	18.5	20.3	16.3	15.7	12.5	11.1	10.7		
		MCWBR	9.2	8.9	9.4	10.7	10.9	10.7	11.1	9.3	9.6	9.4	9.9	10.2		
		MCWBR	9.9	10.4	12.6	15.7	16.6	16.4	18.0	14.4	13.5	11.5	11.2	10.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.4	9.0	9.5	10.3	10.4	10.3	10.7	9.4	9.3	9.5	10.2	10.3		
	taub		0.324	0.338	0.369	0.389	0.389	0.388	0.388	0.373	0.360	0.343	0.318	0.307		
		taud	2.238	2.333	2.279	2.252	2.301	2.334	2.340	2.389	2.436	2.419	2.318	2.214		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		179	229	250	262	270	272	269	266	253	226	183	153		
		Edh at Noon	18	24	32	37	38	37	36	32	27	22	16	15		
	RadAvg		165	368	693	1132	1495	1459	1406	1129	803	438	214	122		
(45) All-Sky Solar Radiation	RadStd		24	35	80	123	104	133	141	84	59	52	21	12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page